

**EVALUATION OF THE AWARENESS AND UTILIZATION OF
ARTIFICIAL INTELLIGENCE
AMONG CIVIL SERVANTS IN SOUTH-WEST NIGERIA**

GROUP 9

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Few developments of recent have altered the nature of professional work as rapidly or as quietly as artificial intelligence. Unlike earlier technological shifts that came with obvious machinery and unmistakable disruption, AI has entered workplaces through applications people already use and even embedded in email clients, search engines, document editors, and communication platforms. The result is that many workers are already interacting with AI tools without always recognising them as such, while others remain very unaware of the capabilities now available to them.

Globally, the rate at which AI has been integrated into professional environments has accelerated sharply since the public release of large language models beginning in late 2022. A study by Ipsos and Google in 2024, conducted across 21 countries found that 48% of respondents globally had used generative AI tools in the preceding year.

Within the Nigerian online population surveyed, that figure rose to 70%, a proportion that significantly exceeded the global average and which reflected a pattern of enthusiastic, sometimes informal, AI adoption among digitally active Nigerians (Ipsos & Google, 2024). Searches in Nigeria for 'how to use AI' reportedly increased by 40 percent during 2024, suggesting that curiosity about these tools is not merely passive (BusinessDay, 2024).

A more recent 2025 global study by KPMG and the University of Melbourne further found that 58% of surveyed employees intentionally use AI for work, while 57% hide their AI use from their colleagues and managers, with 79% of those same organisations maintaining some form of AI governance policy, a figure that reflects how rapidly institutional responses to AI are crystallising in higher-income contexts.

Anecdotal, the name 'ChatGPT' has become broadly synonymous with AI itself across Nigerian workplaces and educational settings, suggesting a level of consumer familiarity that has outpaced any formal instruction or policy response. This pattern of informal, self-directed adoption,

occurring outside institutional frameworks, is particularly relevant to the civil service context that this study examines.

For Nigeria specifically, these trends intersect with a broader national conversation about digital transformation and public sector reform. The federal government has in recent years undertaken a series of initiatives aimed at embedding technology into governance.

The paperless initiative, for instance, expanded from three to thirty-eight federal ministries between 2024 and early 2025, while the Office of the Head of the Civil Service of the Federation introduced ServiceWise GPT, an AI-powered platform trained on government regulations and policy documents that had accumulated over 50,000 interactions by early 2025 and reportedly saved civil servants an estimated two to three hours daily (BusinessDay, 2026; OHCSF, 2025).

In October 2025, the federal government made formal its commitment further by partnering with Google and Apolitical to launch the Government AI Campus, a structured programme designed to deliver AI literacy training to civil servants across ministries, departments, and agencies (Premium Times, 2025).

UNESCO also, through its Framework on AI and Digital Competencies for the Public Sector, trained 80 civil servants from two ministries over five days in February 2025, specifically targeting knowledge gaps in AI governance, data protection, and digital project management (UNESCO, 2025).

These policy-level interventions sit against a background of well-documented structural challenges. Nigeria's National Artificial Intelligence Strategy (NAIS), finalised in 2025 under the Federal Ministry of Communications, Innovation and Digital Economy, explicitly acknowledges that the country faces limited and unreliable infrastructure, low broadband penetration, public research and development investment of approximately 0.2% of GDP against a global average of 2.2%, and a significant shortage of skilled AI professionals compounded by ongoing brain drain (OECD AI Policy Observatory, 2026).

Digital literacy rates, depending on the source and methodology, range from 43-50% of the population (NITDA, 2025). The urban-rural digital divide remains a particularly acute concern, by concentrating whatever AI infrastructure and capacity that exists in major urban centres while leaving smaller cities and rural areas comparatively underserved.

It is within this context of national enthusiasm, policy ambition, and structural constraint that the South-West zone of Nigeria presents itself as a particularly instructive site for inquiry. The region comprising Lagos, Oyo, Ogun, Osun, Ondo, and Ekiti states is home to the country's commercial capital in Lagos, several of its older established universities, and a relatively high concentration of public sector employment.

The civil service in this zone serves millions of Nigerians across a wide range of administrative functions, from revenue collection and urban planning to health service administration and record-keeping. These functions increasingly involve the handling of data, the drafting of policy documents, the coordination of inter-agency correspondence, and the delivery of citizen-facing services, all of which are areas where AI tools might plausibly offer efficiency gains.

Yet despite the growth of both AI usage nationally and policy interest at the federal level, very little is known about how civil servants in South-West Nigeria specifically understand, perceive, and interact with artificial intelligence in their day-to-day work.

It is not clear whether awareness of AI tools extends meaningfully into this workforce, whether those who are aware of these tools are using them in professional contexts, what barriers exist to adoption, or whether institutional frameworks in state and local government establishments are providing any guidance on AI use.

These are not merely academic questions because as AI tools become more deeply embedded in how work gets done globally, public servants who lack awareness of, or access to, these tools risk a growing productivity gap relative to peers in the private sector and in better-resourced government systems elsewhere. The citizens they serve may consequently receive a slower, less accurate, and less responsive form of public administration.

This study therefore sets out to assess the level of awareness and the nature of workplace use of artificial intelligence among civil servants in South-West Nigeria, with the intention of establishing an empirical baseline from which more targeted interventions in training, policy, and institutional design can be planned.

1.2 Statement of Research Problem

The Nigerian civil service, like public administrations across most of sub-Saharan Africa, is navigating a moment of simultaneous opportunity and institutional lag.

On one hand, AI tools have become sufficiently accessible and often requiring nothing more than a smartphone and an internet connection such that their adoption by individual workers is already happening informally and without institutional sanction or support.

A 2023 study on ChatGPT-based models in Nigerian government institutions found strong support among IT personnel for AI integration in public service delivery, but also identified significant concerns around security, employee resistance, and inadequate infrastructure as barriers to systematic deployment (Onwuka et al., 2023).

A 2025 study examining barriers to digital transformation in Lagos State public institutions similarly found that inadequate budget allocation was the single most statistically significant obstacle to digital adoption within government organisations, followed closely by the absence of a clear strategic vision for technology use.

There is also the question of what 'use' means in this context. The 2025 KPMG global study on AI trust and workplace use found that among Nigerian workers surveyed, 44% reported using AI at work in ways that contravene organisational policies and guidelines, 72% reported relying on AI output without evaluating its accuracy, and 69% had presented AI-generated content as entirely their own (KPMG & University of Melbourne, 2025).

These figures suggest that even where AI adoption is occurring, it may be happening without adequate critical literacy, responsible use frameworks, or institutional oversight. In a civil service context, where accuracy, accountability, and confidentiality of government documents are non-negotiable, uninformed or unguided AI use presents its own category of risk.

A comparable dynamic is increasingly observable within the civil service itself, where individual officers have been found using AI tools such as ChatGPT for drafting official correspondence, summarising policy documents, and generating reports, often without any formal organisational sanction or guidance. The use tends to be informal, self-initiated, and undisclosed, driven by personal initiative rather than institutional direction.

This study exists to address precisely that gap. As AI tools become more accessible to civil servants across the South-West, the absence of empirical research documenting how, why, and under what conditions these tools are being adopted within public institutions leaves governments without the evidence base needed to develop informed policy, ethical guidelines, or institutional best-practice frameworks.

This research seeks to surface those adoption behaviours among civil servants, examine their implications for public service delivery, and provide the grounded, data-driven foundation that administrators, institutions, and policymakers urgently need.

1.3 Justification of the Study

The case for this study is grounded in five distinct observations that converge on the same conclusion.

First, AI tools already demonstrate measurable efficiency gains in Nigerian civil service contexts. The OHCSF's ServiceWise GPT data confirms this yet it remains unknown whether such gains are being realised at the state level in South-West Nigeria.

Second, with the IFC projecting that 28 million Nigerian jobs will require digital skills by 2030, workforce readiness is an urgent concern.

Third, uneven AI adoption also risks creating an internal digital divide, producing inconsistent public service quality across ministries and departments.

Fourth, existing research has concentrated heavily on federal institutions and Lagos-based tech respondents, leaving South-West state civil services significantly underexplored. The TAM and KAP frameworks, though proven productive in Ghana and Tanzania, remain largely unapplied to the Nigerian context.

Finally, having observed firsthand the rapid and unregulated integration of AI tools into professional practice across various sectors, it is striking how little of this reality is reflected in any formal government or institutional response at the state level.

1.4 Aim and Objectives of the Study

The broad aim of this study is to assess the level of awareness and workplace utilization of Artificial Intelligence (AI) among civil servants in South-West Nigeria. Specifically, the study seeks to:

- i. Determine the current level of awareness and knowledge regarding AI technologies among civil servants in South-West Nigeria.
- ii. Identify the specific AI tools and applications currently being utilized by civil servants in the execution of their professional duties.
- iii. Evaluate the perceived impact of AI adoption on workplace productivity, operational efficiency, and public service delivery.
- iv. Identify the structural, institutional, and personal barriers limiting the effective adoption and integration of AI tools within the state civil service.

1.5 Research Questions

To guide this study and achieve the stated objectives, the following research questions have been formulated:

- i. What is the current level of awareness and knowledge of AI technologies among civil servants in South-West Nigeria?
- ii. Which specific AI tools and applications are predominantly used by civil servants in their day-to-day workplace operations?
- iii. How does the utilization of AI tools influence the perceived efficiency, productivity, and service delivery of the civil servants?
- iv. What are the major structural, institutional, and personal barriers restricting the adoption of AI in the South-West Nigerian civil service?

1.6 Scope of the Study

This study is geographically delimited to the South-West geopolitical zone of Nigeria, encompassing Lagos, Ogun, Oyo, Osun, Ondo, and Ekiti states. Conceptually, the research focuses exclusively on artificial intelligence awareness, patterns of workplace usage, perceived productivity impacts, and adoption barriers. The demographic scope is strictly restricted to

active, formally employed civil servants operating within state government Ministries, Departments, and Agencies (MDAs). Consequently, private sector employees, informal workers, retired public servants, and federal civil servants not directly integrated into the state bureaucratic machinery are excluded from the primary sample.

1.7 Operational Definition of Terms

Artificial Intelligence (AI): In the context of this study, AI refers to digital tools, software, and platforms (such as generative language models, automated document processors, and data analytics tools) used to automate tasks, generate content, or enhance administrative decision-making.

Civil Servant: An individual formally employed in the public administration sector of the state governments in South-West Nigeria, working within established Ministries, Departments, and Agencies (MDAs).

Workplace Use: The integration and practical application of AI technologies by employees to execute their official duties, whether these tools are formally sanctioned by the institution or adopted informally on an individual basis.

Awareness: The baseline knowledge, recognition, and understanding of AI tools, their capabilities, and their potential applications within a public service environment.

Service Delivery: The systematic process and efficiency with which government policies, administrative duties, and public services are executed and provided to citizens.

Digital Divide: The gap between civil servants who have access to, and knowledge of, modern digital technologies (including AI) and those who do not, often influenced by infrastructure, training, or geographic location.

1.8 Organization of Report

This report is structured into five chapters to ensure a logical flow and clarity of presentation. Chapter One introduces the study, outlining the background, problem statement, objectives, research questions, and significance. Chapter Two presents a review of relevant literature, including theoretical frameworks and previous studies on AI in public administration. Chapter Three details the research methodology, covering the study design, population, sampling

techniques, data collection instruments, and analysis methods. Chapter Four presents the results and analysis of the survey data, integrating both quantitative and qualitative findings. Finally, Chapter Five provides a discussion of key findings, conclusions drawn from the study, and practical recommendations for improving AI awareness and implementation among civil servants in South-West Nigeria.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews relevant conceptual, theoretical, and empirical literature on the awareness and workplace use of Artificial Intelligence (AI) among civil servants, with particular attention to the South-West geopolitical zone of Nigeria. The chapter is organised into four broad sections. The first section undertakes a conceptual review of the key terms central to the study, namely artificial intelligence, awareness, workplace use of AI, and the civil service system in South-West Nigeria. The second section examines the theoretical frameworks that corroborates technology acceptance and adoption research, namely the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Diffusion of Innovation (DOI) theory, and identifies the framework adopted for the present study. The third section reviews empirical studies conducted globally, across Africa, and within Nigeria, with emphasis on studies that have examined AI awareness and adoption in public administration. The final section identifies gaps in the existing literature that the present study seeks to address, and closes with a summary of the chapter.

2.2 Conceptual Review

2.2.1 Concept of Artificial Intelligence

Artificial Intelligence (AI) refers broadly to the capability of machines and computer systems to perform tasks that typically require human intelligence, such as reasoning, learning, problem-solving, perception, and decision-making. The term traces its intellectual origin to the 1956 Dartmouth Summer Research Project, which is widely regarded as the formal starting point of AI as a field of scientific inquiry (McCarthy, Minsky, Rochester, & Shannon, 2006, as cited in Nwosu, Obalum, & Ananti, 2024).

In the context of public administration, AI is generally understood as the application of these intelligent systems to government functions such as service delivery, policy analysis, fraud detection, record management, citizen engagement, and administrative decision-making (Young, Bullock, & Lecy, 2019, as cited in Nwosu et al., 2024). Scholars have noted that AI in

government is not a single monolithic technology but a spectrum of tools ranging from simple rule-based automation and chatbots to more advanced machine-learning-driven analytics and predictive systems (Reis, Santo, & Melao, 2019, as cited in Nwosu et al., 2024). For the purposes of this study, AI is conceptualised as the range of digital tools and applications, including generative AI chatbots, automated document and data-processing systems, AI-enabled e-government platforms, and decision-support software, that civil servants may encounter or use in the course of their official duties.

2.2.2 Concept of Awareness and AI Awareness

Awareness, in the context of technology adoption, refers to an individual's knowledge of the existence, features, and potential applications of a given innovation prior to forming an attitude towards it or deciding whether to use it. Within Rogers' classical formulation of the innovation-decision process, awareness (or knowledge) constitutes the first stage through which a potential adopter becomes exposed to an innovation and begins to understand how it functions, a stage that necessarily precedes persuasion, decision, implementation, and confirmation. Applied to AI, awareness denotes the extent to which an individual, in this case a civil servant, knows that AI tools exist, understands in broad terms what such tools can do, and recognises possible applications of AI within their own work environment.

2.2.3 Workplace Use of AI in Public Administration

Workplace use of AI refers to the actual, observable deployment of AI-enabled tools by employees to perform job-related functions, as distinct from mere awareness or favourable attitudes towards the technology. In public administration specifically, workplace use of AI has been documented across several functional areas, including automated correspondence and document drafting, data analytics for policy formulation, chatbot-based citizen service platforms, predictive systems for resource allocation, and administrative process automation (Sharma, Yadav, & Chopra, 2020, as cited in Nwosu et al., 2024). Globally, governments have piloted AI-driven systems to improve efficiency, reduce administrative bottlenecks, and enhance transparency in service delivery, although the scale and sophistication of such applications vary considerably between developed and developing economies (Agba, Agba, & Obeten, 2023, as cited in Nwosu et al., 2024).

2.2.4 Overview of the Civil Service System in South-West Nigeria

The civil service in Nigeria's South-West geopolitical zone, comprising Lagos, Ogun, Oyo, Osun, Ondo, and Ekiti States, constitutes the administrative machinery responsible for implementing government policies and delivering public services at the state level. Lagos State, as Nigeria's commercial nerve centre, has historically been at the forefront of digital governance reforms within the zone, having invested in e-governance platforms, online revenue collection systems, and digital service portals (Hassan, Fatile, & Ashade, 2023, as cited in Nwosu et al., 2024). Other states within the zone have made comparatively slower progress in digital transformation, often constrained by funding limitations, infrastructural deficits, and capacity gaps among personnel.

The South-West zone is nonetheless frequently identified in the literature as a relatively favourable environment for technology-driven public-sector reform, owing to its higher concentration of tertiary institutions, technology hubs, and private-sector digital innovation activity compared with other regions of the country (PwC Nigeria, Lagos Business School, & Microsoft, 2025). This relatively enabling environment makes the zone a useful setting in which to examine how far AI awareness and workplace use have actually permeated routine civil service practice, since favourable contextual conditions do not automatically translate into widespread awareness or use at the level of individual employees.

2.3 Theoretical Framework

Several theories have been advanced in the information systems and innovation-diffusion literature to explain how and why individuals come to know about, accept, and use new technologies. Three of the most widely applied frameworks, the Technology Acceptance Model, the Unified Theory of Acceptance and Use of Technology, and the Diffusion of Innovation theory, are reviewed below.

2.3.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model, originally developed by Davis (1989), posits that an individual's intention to use a new technology is primarily determined by two beliefs: perceived usefulness, the degree to which a person believes that using the technology will enhance their job performance, and perceived ease of use, the degree to which a person believes that using the technology will be free of effort. These two beliefs shape attitude towards use, which in turn influences behavioural intention and actual system use. In the context of AI in public services,

TAM has been applied to assess how public servants perceive the usefulness and ease of use of AI systems, and how these perceptions shape their willingness to adopt such systems in practice (Nwosu et al., 2024). The model has also been combined with the Technology-Organisation-Environment (TOE) framework in studies of government AI acceptance, with findings indicating that institutional and organisational readiness factors significantly shape individual perceptions of usefulness and ease of use of AI tools within government ministries.

2.3.2 Unified Theory of Acceptance and Use of Technology (UTAUT)

The Unified Theory of Acceptance and Use of Technology, developed by Venkatesh, Morris, Davis, and Davis (2003), extends and consolidates earlier technology-acceptance models by proposing four core determinants of behavioural intention and use behaviour: performance expectancy, effort expectancy, social influence, and facilitating conditions, with gender, age, experience, and voluntariness of use serving as moderating variables. UTAUT has been widely applied in Nigerian technology-adoption research, including studies of e-government and e-agriculture adoption, where performance expectancy, effort expectancy, and social influence have consistently emerged as significant predictors of behavioural intention to adopt new digital tools. Applied to AI in the civil service, UTAUT suggests that civil servants are more likely to use AI tools at work when they believe such tools will improve their job performance, when the tools are perceived as easy to learn and use, when supervisors and peers endorse their use, and when adequate organisational and technical support is available.

2.3.3 Diffusion of Innovation (DOI) Theory

Rogers' Diffusion of Innovation theory explains how an innovation is communicated over time among members of a social system, and identifies five stages of the innovation-decision process: knowledge (awareness), persuasion, decision, implementation, and confirmation. The theory further classifies adopters into five categories, namely innovators, early adopters, early majority, late majority, and laggards, based on the relative timing of their adoption decisions. DOI theory is particularly relevant to the present study because it explicitly foregrounds awareness, or 'knowledge,' as the necessary first stage that precedes any adoption decision, making it well suited to a study that examines awareness as a distinct construct from actual workplace use.

2.3.4 Theoretical Framework Adopted for the Study

This study draws on an integration of the Diffusion of Innovation theory and the Technology Acceptance Model. DOI theory provides the conceptual basis for distinguishing awareness (knowledge) from subsequent adoption-related behaviour, which aligns directly with the study's dual focus on awareness and workplace use of AI. TAM complements this by explaining the cognitive mechanisms, namely perceived usefulness and perceived ease of use, through which awareness is translated, or fails to be translated, into actual workplace use among civil servants. Elements of UTAUT, particularly social influence and facilitating conditions, are also drawn upon to account for the organisational and institutional factors that may enable or constrain AI use within South-West Nigeria's civil service environment.

2.4 Empirical Review

2.4.1 Global and Cross-National Studies on AI Awareness and Adoption in the Public Sector

A growing body of international research has examined the factors that shape public-sector employees' acceptance and use of AI. Ahn and Chen (2022), in a widely cited study on AI-augmented public administration, found that government employees' willingness to use AI was strongly associated with their perception of organisational readiness and their existing digital literacy, with employees who had prior exposure to digital tools reporting greater willingness to adopt AI-based systems. Similarly, Alshahrani, Griva, Dennehy, and Mantymaki (2024), reviewing AI-supported decision-making in government functions, identified data governance, algorithmic transparency, and workforce capability as recurring themes shaping the opportunities for, and challenges of, AI adoption across public institutions internationally.

Research situated in other regions has produced broadly consistent findings. A study of generative AI adoption among employees in South Korean companies, applying the UTAUT model, found that effort expectancy and social influence were the strongest predictors of behavioural intention to use generative AI systems, with effort expectancy more influential at early adoption stages and social influence, particularly support from supervisors and peers, more influential as adoption proceeded; performance expectancy and facilitating conditions, by contrast, showed weaker or non-significant effects in that context. Other TAM-based studies of AI acceptance, including research applying TAM to AI use in e-commerce settings, similarly emphasise that perceived usefulness and ease of use, together with users' attitudes and concerns

about AI, are critical to understanding why some categories of users adopt AI tools readily while others remain hesitant. Within government settings specifically, an empirical study assessing the acceptance of AI technologies in a governmental sector, combining the Technology-Organisation-Environment framework with TAM, found that institutional factors such as organisational readiness and the legal framework significantly shaped personnel's perceived usefulness and ease of use of AI, reinforcing the view that individual-level technology acceptance in government cannot be fully understood without reference to organisational context.

2.4.2 AI Awareness and Adoption Studies in the Nigerian Public Service

Within the Nigerian context, several recent studies have begun to interrogate the state of AI awareness and adoption in public administration. Nwosu, Obalum, and Ananti (2024), in a qualitative study based on secondary data, examined the current state of AI implementation in Nigeria's public service and found that AI adoption remains at an early stage, with emerging applications limited mainly to e-governance, healthcare, banking, real estate, and security services. Their study identified inadequate infrastructure, limited digital literacy, insufficient human capital, and weak public awareness of AI as the major barriers constraining wider adoption, and recommended significant investment in infrastructure and human-capital development, alongside mandatory training programmes for public servants, to close existing skill gaps.

In a related contribution focusing specifically on recent graduates, Ugwuozor and Egenti (2024), as cited in Nwosu et al. (2024), surveyed 112 graduates undergoing the National Youth Service Corps and used descriptive statistics and a multinomial logistic model to assess their awareness of AI and their preparedness for the opportunities and threats AI may pose to their future work life; the study highlighted a generally positive but cautious outlook among young, educated Nigerians, alongside an acknowledgement of significant skill gaps. At the policy level, Obi, Ole, and Uzoigwe (2023), as cited in Nwosu et al. (2024), examined the use of AI systems in Nigeria and the trajectory of AI policy development, underscoring the absence, until relatively recently, of a comprehensive national regulatory framework to guide AI deployment, including within public institutions.

Broader continental research situates the Nigerian experience within the wider African pattern of AI adoption. Gwagwa, Kraemer-Mbula, Rizk, Rutenberg, and De Beer (2020), as cited in Nwosu et al. (2024), analysed AI deployments across Africa and identified policy, infrastructural, and skills-related dimensions that shape the benefits and challenges of AI uptake on the continent, while Plantinga (2024), as cited in Nwosu et al. (2024), examined digital discretion in African public administration and its implications for AI use, cautioning that the introduction of AI tools into administrative decision-making can alter the discretionary latitude traditionally exercised by street-level bureaucrats, with consequences for accountability and service quality that are not yet well understood in African settings, including Nigeria.

More recent developments suggest that government-led capacity-building efforts are beginning to address the awareness gap identified in earlier studies. In 2025 and 2026, the Nigerian Federal Government, in partnership with UNESCO, commenced AI-literacy training programmes for senior civil servants, covering topics such as open data and open government, fundamentals of AI, data protection and privacy, and digital-transformation strategy, with the explicit goal of strengthening digital governance and promoting ethical AI adoption across government institutions (Vanguard News, 2026; Voice of Nigeria, 2026). UNESCO's earlier 2021 continental survey had found that the great majority of African countries, including Nigeria, desired external support for AI capacity-building among civil servants, underscoring that the awareness deficit was a continent-wide, rather than purely national, concern (UNESCO, 2025). Separately, the Federal Ministry of Communications and Digital Economy, in partnership with Google and Apolitical, launched the AI Government Campus initiative to provide structured online AI training to civil servants across ministries, departments, and agencies, as part of a broader effort to embed AI capability natively within the civil service (iAfrica.com, 2025; Premium Times, 2025).

2.4.3 Studies on AI in South-West Nigeria and Lagos State

Empirical attention to AI specifically within South-West Nigeria remains comparatively limited, though the available evidence points to Lagos State as the most advanced sub-national jurisdiction in the zone with respect to digital governance. Hassan, Fatile, and Ashade (2023), as cited in Nwosu et al. (2024), undertook an assessment of artificial intelligence in public administration with specific reference to implications for service delivery in the Lagos State

public service, a study that remains one of the few empirical works to examine AI-related practice at the sub-national level in South-West Nigeria. Complementing this, broader analyses of Lagos's digital transformation trajectory have observed that, notwithstanding the state's relative advantages in infrastructure and private-sector technology activity, AI implementation within Lagos's public institutions is still at an embryonic stage, with recommendations for greater collaboration between state government, the private sector, and financial institutions to accelerate broadband penetration and digital transformation (Nwosu et al., 2024).

A joint industry whitepaper produced by PwC Nigeria, Lagos Business School, and Microsoft similarly observes that, while AI remains an emerging field in Nigeria generally, recent years have witnessed a notable shift in awareness, investment, and application, propelled in part by entrepreneurs, researchers, and policymakers based in Lagos and other South-West technology hubs (PwC Nigeria, Lagos Business School, & Microsoft, 2025). Taken together, these contributions suggest that while South-West Nigeria, and Lagos State in particular, may possess more favourable structural conditions for AI awareness and use than other parts of the country, rigorous, employee-level empirical evidence on actual awareness and workplace use of AI among civil servants across the zone's various states remains scarce, a gap that the present study seeks to address.

2.5 Gaps in the Literature

The foregoing review reveals several gaps in the existing body of knowledge. First, much of the available Nigerian literature on AI in public administration relies on qualitative or secondary-data designs focused on national-level policy and infrastructure questions, with comparatively few studies generating primary, employee-level survey data on individual civil servants' awareness and actual workplace use of AI. Second, where empirical attention has been paid to sub-national contexts, it has tended to concentrate on Lagos State, leaving the awareness and usage patterns of civil servants in Ogun, Oyo, Osun, Ondo, and Ekiti States comparatively under-examined, despite these states forming part of the same geopolitical zone and sharing broadly similar administrative structures. Third, few studies have explicitly distinguished between awareness of AI and actual workplace use of AI as separate, measurable constructs, despite theoretical models such as DOI theory indicating that awareness is a necessary, but not sufficient, precondition for use; this conflation makes it difficult to determine whether low AI

utilisation in the public service stems from a lack of awareness, unfavourable attitudes despite awareness, or structural and organisational barriers to use. Finally, although recent government-led training initiatives (such as the UNESCO AI-literacy programme and the AI Government Campus) suggest growing institutional commitment to building AI capacity within the civil service, independent empirical evaluation of the actual reach, effectiveness, and on-the-ground impact of these initiatives on rank-and-file civil servants' awareness and use of AI remains largely absent from the scholarly literature. The present study addresses these gaps by generating primary, employee-level evidence on AI awareness and workplace use across civil servants in South-West Nigeria.

2.6 Summary of the Chapter

This chapter has reviewed conceptual, theoretical, and empirical literature relevant to the study of AI awareness and workplace use among civil servants in South-West Nigeria. The conceptual review clarified the meaning of artificial intelligence, awareness, and workplace use of AI, and situated these concepts within the structure of the civil service system in South-West Nigeria. The theoretical review examined the Technology Acceptance Model, the Unified Theory of Acceptance and Use of Technology, and the Diffusion of Innovation theory, and outlined an integrated DOI-TAM framework, supplemented by elements of UTAUT, as the theoretical lens adopted for the study. The empirical review showed that, while global and cross-national research has generated relatively rich evidence on the determinants of AI acceptance and use among public-sector employees, Nigerian and, more specifically, South-West Nigerian literature on this subject remains comparatively thin, qualitative in orientation, and concentrated on Lagos State. These observations confirm the need for the present study, which seeks to generate primary, comparative, employee-level evidence on AI awareness and workplace use among civil servants across South-West Nigeria.

CHAPTER THREE

METHODOLOGY

3.1 Research Design

This study employed a descriptive, cross-sectional survey design to evaluate the level of awareness, understanding, and workplace utilization of Artificial Intelligence (AI) tools among civil servants in South-West Nigeria. The research utilized an online questionnaire developed using Google Forms, enabling efficient data collection from participants across different professional categories and geographic locations within the six South-West states.

The survey was structured based on two theoretical frameworks:

- i. The Diffusion of Innovation (DOI) theory was employed to assess participants' awareness of AI (knowledge stage), their attitudes towards its relevance (persuasion stage), and their reported or intended use of AI tools in official duties (implementation and confirmation stages).
- ii. The Technology Acceptance Model (TAM) was also adapted to explore perceptions of AI's usefulness and ease of use in civil service work, both of which are predictors of behavioural intention to adopt.

This mixed-framework approach allowed for a nuanced understanding of not only what civil servants know about AI, but also how likely they are to adopt and sustain its use in professional practice.

3.2 Population and Sampling

3.2.1 Target Population

The target population for this study comprised active, formally employed civil servants working within state government Ministries, Departments, and Agencies (MDAs) across the six states of South-West Nigeria: Lagos, Ogun, Oyo, Osun, Ondo, and Ekiti. Given the nature of AI adoption and its applicability to a wide range of administrative functions, the study included civil servants across cadre classifications, from secretarial and clerical staff to administrative, professional, and technical personnel.

3.2.2 Sampling Technique

A combination of convenience and snowball sampling methods was employed. The survey was initially distributed through the research team's existing professional networks, including WhatsApp groups of civil servants, government agency alumni networks, and colleague referrals. Recipients were encouraged to forward the survey link to eligible colleagues within their MDAs, facilitating peer-referral recruitment to extend the reach of the sample beyond the immediate networks of the research team.

3.2.3 Sample Size

The study collected a total of 171 responses from civil servants across the six South-West states. Of these, 161 were deemed valid and included in the final analysis after data cleaning, representing a data retention rate of 94.2%. While the sample size represents a convenience-based rather than probability-drawn sample, it provides a diverse and meaningful cross-section of civil servant cadres, experience levels, age groups, and states. The findings, while not statistically generalizable to the entire South-West civil service population, offer a valuable empirical baseline for understanding AI awareness and adoption patterns within this workforce.

Inclusion Criteria: Respondents who were actively employed as civil servants in a state government MDA within the South-West geopolitical zone of Nigeria, and who voluntarily consented to participate.

Exclusion Criteria: Responses were excluded if the individual was not a current state civil servant in the South-West zone, was based outside the zone, or did not provide informed consent.

3.3 Data Collection Method

A cross-sectional survey design was employed using a Google Forms questionnaire to collect data on AI awareness, tool familiarity, adoption status, perceived usefulness, ease of use, and institutional barriers among civil servants. The questionnaire began with an introductory page outlining the study's purpose, expected duration, and how data would be used, with an explicit statement of confidentiality and voluntary participation. Respondents were required to acknowledge informed consent before proceeding.

No personal identifiers (such as names, exact workplace location, or personal contact details) were requested. The following demographic variables were collected:

- i. State of employment
- ii. Cadre or rank classification
- iii. Years of experience in the civil service
- iv. Age range
- v. Highest educational qualification

The questionnaire combined multiple-choice, yes/no, Likert-scale, multi-select, and open-ended questions, structured to progressively move from general awareness to specific adoption behaviours and institutional readiness.

3.4 Survey Development and Administration

3.4.1 Structure of the Questionnaire

The final instrument was structured into thematic sections aligned with the study's objectives:

- i. Demographic Information: Background data on respondents' state, cadre, experience, age, and qualification.
- ii. AI Awareness and Familiarity: Items exploring whether civil servants had heard of AI, their self-rated understanding, and familiarity with specific tools.
- iii. AI Tool Adoption and Usage: Questions on current adoption status, the tasks for which AI is used, and the primary device used for access.
- iv. Technology Acceptance Model (TAM) Constructs: Six Likert-scale items measuring Perceived Usefulness and Perceived Ease of Use.
- v. Challenges and Barriers: Multi-select question on workplace challenges limiting AI use.
- vi. Training and Institutional Readiness: Items on formal AI training received and willingness to participate in future training.
- vii. Government Priorities: An open-ended question inviting respondents to identify what the government should prioritise to improve technology adoption.

3.4.2 Survey Platform and Distribution

The questionnaire was built and hosted on Google Forms. Google Forms was selected for its no-cost access and reliability, mobile responsiveness and accessibility, and easy integration with cloud-based data export. The survey was administered over approximately two weeks in June 2026. The survey link was disseminated via WhatsApp groups for civil servants, professional networks, and personal contact chains within government offices. Recipients were encouraged to forward the survey link to eligible colleagues, facilitating a snowball expansion of the sample.

Before public dissemination, the form was tested internally across different devices (phones, tablets, laptops) to ensure cross-platform functionality.

3.4.3 Response Rates and Mitigation Measures

Initial response rates were modest, reflecting a common limitation of online surveys distributed to working professionals. Many respondents cited workload pressures and heavy administrative schedules as reasons for delayed participation. To address this, reminders were sent through the same channels, and group members engaged in personal follow-ups with their professional networks. By the end of the survey period, 171 total responses had been received, with 161 retained after data cleaning.

3.5 Ethical Considerations

The survey procedures conformed to accepted research ethics and data protection standards. Informed consent was obtained in advance: the initial survey page clearly explained the study objectives, usage of data, and that participation was voluntary. Respondents could freely choose to participate or not, with no penalty for declining. The consent information also informed participants of their right to skip any questions or withdraw at any point without penalty.

Anonymity and confidentiality were central ethical safeguards. No direct identifiers were captured, which reduces the risk of privacy breaches and encourages candour. All survey responses were collected anonymously and stored securely within Google Forms, with access restricted to the research team. No personal identifiers were recorded. Raw data was retained only for the duration of the study and will be securely deleted or archived according to institutional policy.

3.6 Challenges Encountered

Despite careful planning and preparation, the research team encountered a number of challenges in the course of conducting this study:

- i. **Uneven geographic representation:** The snowball and convenience sampling approach produced uneven representation across the six South-West states. Ogun State was substantially underrepresented in the final sample, with only 2 respondents, reflecting the research team's limited professional network in that state.
- ii. **Non-response and late responses:** As is typical of online surveys distributed to working professionals, initial response rates were modest. Many civil servants cited workload pressures and heavy administrative schedules as reasons for delayed participation.
- iii. **Self-selection bias:** Participation in the survey was voluntary and dependent on self-selection. Civil servants with greater familiarity with digital tools or greater interest in AI-related topics may have been disproportionately likely to complete the survey, potentially upwardly biasing estimates of AI awareness and use.
- iv. **Absence of a formal pilot study:** Owing to time constraints, no formal external pilot test of the survey instrument was conducted with a separate sample of civil servants prior to full deployment.
- v. **Verification of civil service status:** Because the survey was distributed through open digital channels, it was not possible to independently verify that all respondents were current civil servants in the South-West zone.

These challenges are consistent with those frequently reported in online survey research targeting busy professional populations in developing-country contexts. They are documented here to support transparent interpretation of the study's findings and to inform the design of future research in this domain.

CHAPTER FOUR

DATA ANALYSIS AND FINDINGS

4.1 Overview

This chapter presents a comprehensive analysis of data collected from 161 valid responses obtained from civil servants across the South-West states of Nigeria through an online Google Forms survey conducted in June 2026. The survey was designed to evaluate the awareness, and adoption of Artificial Intelligence (AI) tools among civil servants within Ministries, Departments, and Agencies (MDAs). A total of 171 responses were received; 161 were deemed valid and included in the final analysis after preliminary data cleaning.

The analysis addresses the study's key research objectives: assessing the level of AI awareness among civil servants, evaluating current AI tool adoption and usage patterns, examining perceptions of usefulness and ease of use through the Technology Acceptance Model (TAM) constructs, identifying barriers to adoption, and exploring institutional readiness with respect to training and capacity building.

Descriptive statistics, including frequencies, percentages, and means, are used to summarise the data. Inferential techniques including chi-square tests, Pearson correlation coefficients, and one-way ANOVA are applied to explore relationships and group differences. Open-ended responses are thematically coded to supplement quantitative findings. Visual representations accompany each section to aid interpretation. The findings provide critical insights into the current state of AI adoption within the Nigerian civil service, identifying key enablers, gaps, and recommendations for improvement.

4.2 Data Preparation and Quality Assessment

4.2.1 Response Rate and Data Completeness

The survey received a total of 171 responses over the data collection window. After excluding 10 incomplete or duplicate entries, 161 responses were retained for analysis, representing a data retention rate of 94.2%. All core variables recorded a completeness rate exceeding 85%, with the open-ended section on government priorities (Section 22) having the highest rate of

non-response (approximately 22 missing values), which is typical for optional qualitative questions in online surveys.

The survey link was disseminated through professional WhatsApp groups, and friends and family who are civil servants, reaching civil servants across six South-West states: Lagos, Oyo, Osun, Ondo, Ekiti, and Ogun. The broad distribution strategy resulted in a geographically diverse and professionally varied sample, lending credibility to the findings.

4.2.2 Data Processing and Variable Creation

Data downloaded from Google Forms was imported into Python (pandas library) for cleaning and analysis. Several derived variables were created to facilitate analysis. A Likert Score (1-5) was computed for each of the six Technology Acceptance Model (TAM) statements by mapping response categories (Strongly Disagree = 1 to Strongly Agree = 5) to numerical values. Multi-select responses for AI tool familiarity, task usage, and workplace challenges were parsed by keyword matching to extract individual category frequencies. An AI Understanding Score was treated as a continuous variable (1-5) based on respondent self-rating. All statistical tests were performed using Python's scipy.stats library, with a significance threshold of $p < 0.05$.

4.3 Demographic Profile of Respondents

4.3.1 State Distribution

Respondents were drawn from all six South-West states of Nigeria. Osun State contributed the largest share with 49 respondents (28.7%), closely followed by Lagos State with 47 respondents (27.5%). Ondo State accounted for 26 respondents (15.2%), Oyo State for 24 (14.0%), Ekiti for 13 (7.6%), and Ogun State for 2 respondents (1.2%). The dominance of Osun and Lagos is likely attributable to the survey distribution strategy, which was anchored in professional networks linked to those states. Despite this skew, the data provides meaningful representation from across the South-West region.

Figure 4.1: Distribution of Respondents by State

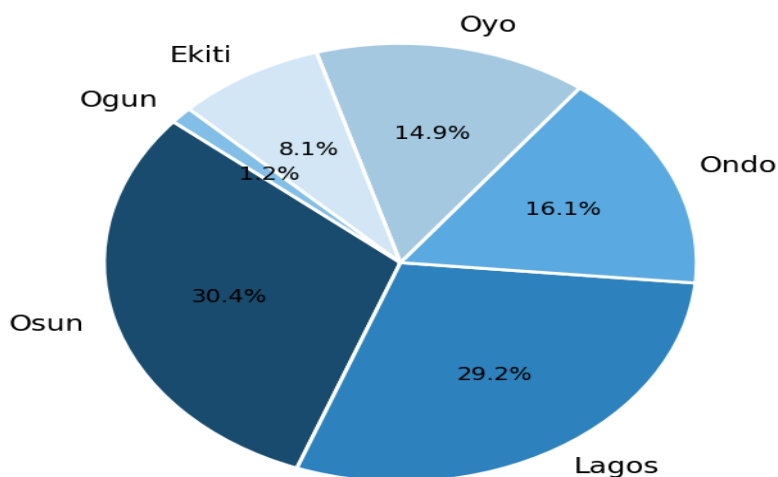


Figure 4.1: Distribution of Respondents by State

4.3.2 Cadre/Rank Classification

The Administrative/Executive cadre constituted the largest professional group in the sample, comprising 74 respondents (43.3%). The Professional/Technical cadre followed with 49 respondents (28.7%), while the Secretarial/Clerical cadre accounted for 23 respondents (13.5%). The remaining 15 respondents (8.8%) identified as 'Other,' including supervisors and mid-level managers not fitting standard classifications.

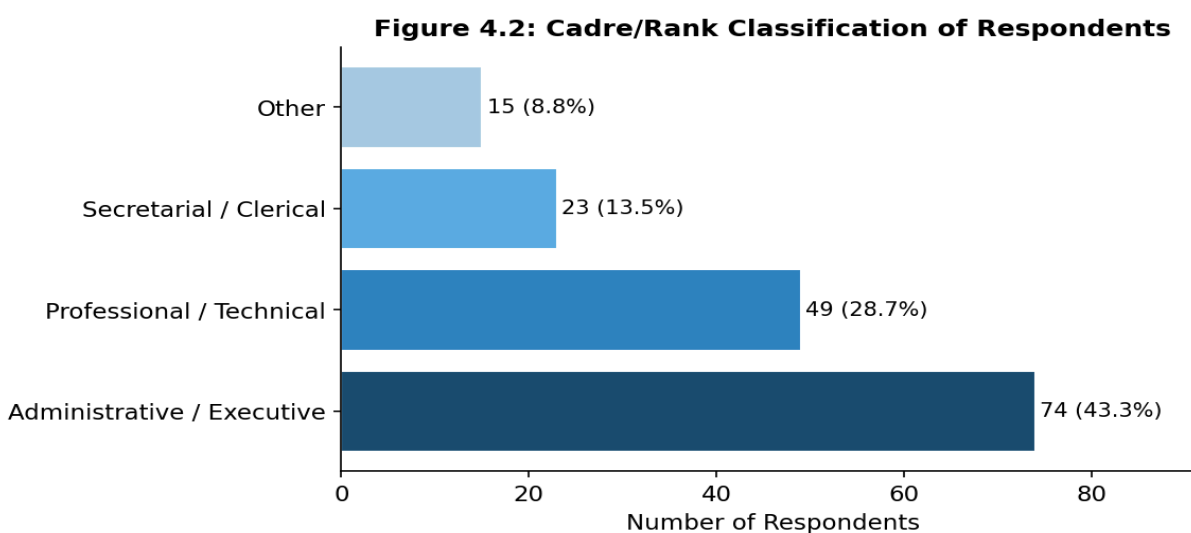


Figure 4.2: Cadre/Rank Classification of Respondents

4.3.3 Years of Experience

The sample was fairly distributed across experience levels, with two experience bands tied at the top: 'Less than 5 years' and 'Above 20 years' each accounted for 43 respondents (25.1%) respectively. Those with '11 to 20 years' of service represented 40 respondents (23.4%), and those in the '5 to 10 years' band numbered 33 (19.3%). The bimodal distribution suggests a generational divide in the dataset that may be relevant when interpreting AI familiarity and adoption patterns.

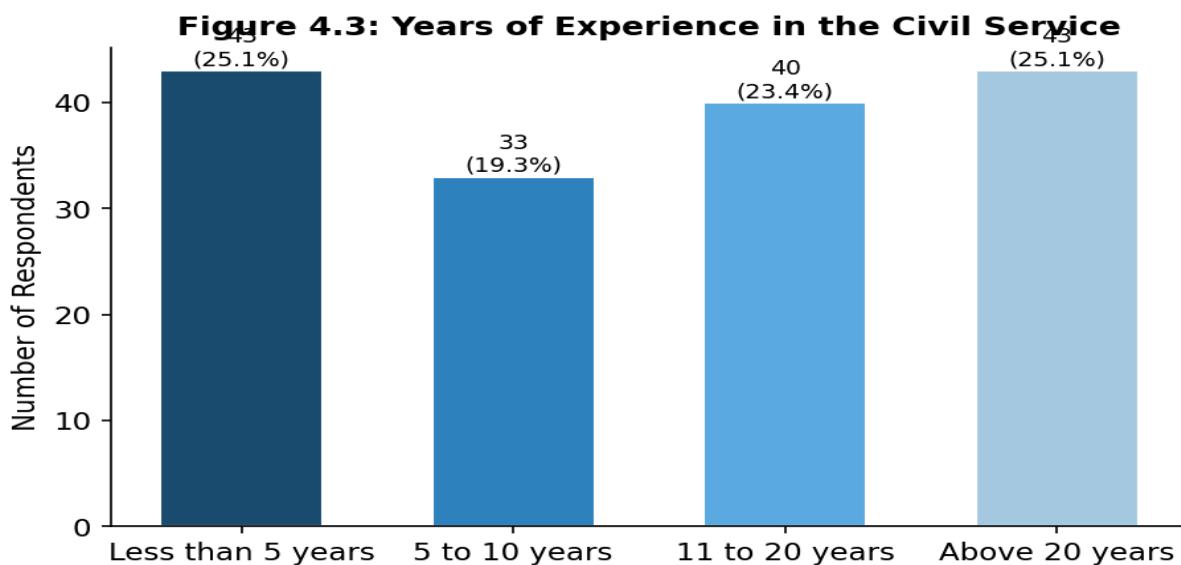


Figure 4.3: Years of Experience in the Civil Service

4.3.4 Age Range

The majority of respondents fell within the 35 to 44 years age bracket, accounting for 52 individuals (30.4%), followed by the 45 to 54 years bracket with 46 respondents (26.9%). Respondents aged 25 to 34 numbered 25 (14.6%), while those aged 55 and above represented 24 respondents (14.0%). The youngest cohort, those below 25 years, was the smallest group with 15 respondents (8.8%).

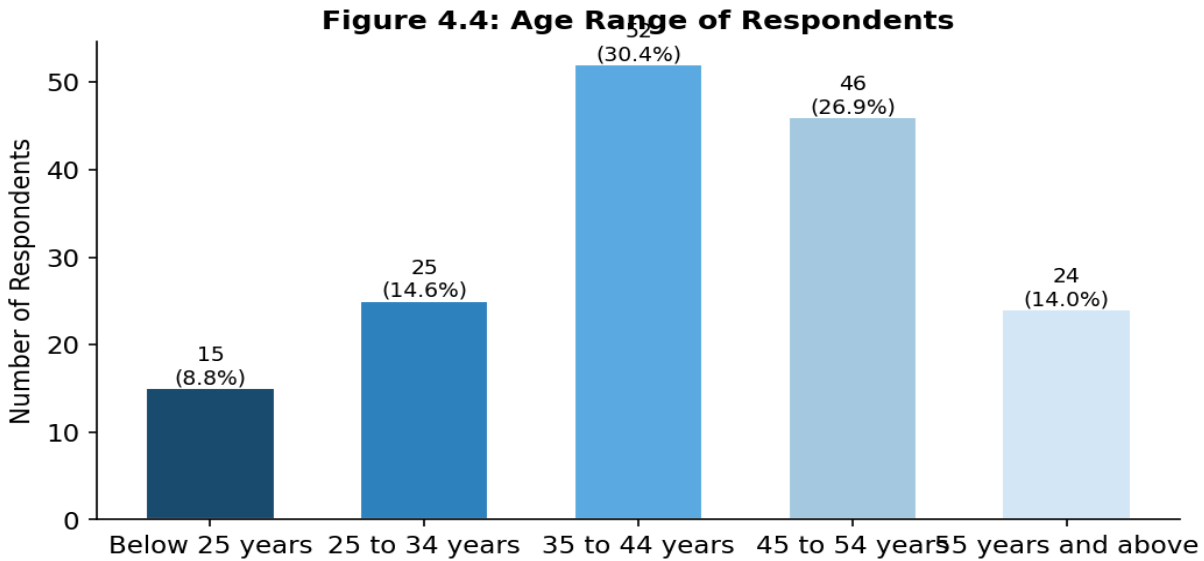


Figure 4.4: Age Range of Respondents

4.3.5 Educational Qualification

A substantial majority of respondents hold undergraduate-level qualifications. HND/Bachelor's Degree holders constituted the largest group at 81 respondents (47.4%), followed by those with a Master's Degree (34 respondents, 19.9%) and OND/NCE holders (27 respondents, 15.8%). PhD holders and those in the 'Other' category each contributed 10 respondents (5.8% each).

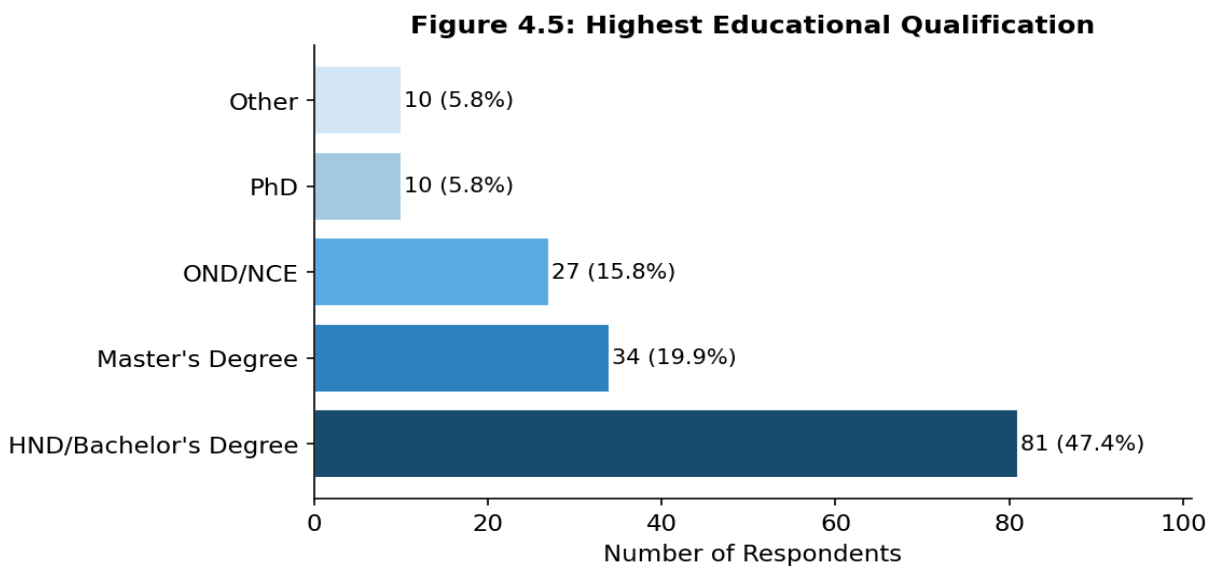


Figure 4.5: Highest Educational Qualification of Respondents

Table 4.1 below provides a statistical summary of the demographic profile across the full sample.

Demographic Variable	Category	Frequency (n)	Percentage (%)
State	Osun	49	28.7%
	Lagos	47	27.5%
	Ondo	26	15.2%
	Oyo	24	14.0%
	Ekiti	13	7.6%
	Ogun	2	1.2%
Cadre	Administrative/Executive	74	43.3%
	Professional/Technical	49	28.7%
	Secretarial/Clerical	23	13.5%
	Other	15	8.8%
Experience	Less than 5 years	43	25.1%
	5–10 years	33	19.3%
	11–20 years	40	23.4%
	Above 20 years	43	25.1%
Qualification	HND/Bachelor's	81	47.4%
	Master's Degree	34	19.9%
	OND/NCE	27	15.8%
	PhD	10	5.8%
	Other	10	5.8%

Table 4.1: Demographic Summary of Respondents (N = 161 valid responses from 171 total)

4.4 AI Awareness and Familiarity

4.4.1 Prior Awareness of Artificial Intelligence

The vast majority of respondents (149 out of 161, or 92.5%) indicated that they had heard of the term Artificial Intelligence prior to participating in the survey. Only 12 respondents (7.5%) reported having no prior awareness of the concept. This high level of general awareness is encouraging and suggests that AI as a concept has permeated the consciousness of the civil service workforce, likely through mass media, social networks, and general digital exposure.

However, as explored in subsequent sections, prior exposure to the term does not necessarily translate into operational knowledge or confident adoption. This distinction between nominal awareness and functional understanding is a central finding of this analysis.

Figure 4.6: Prior Awareness of Artificial Intelligence

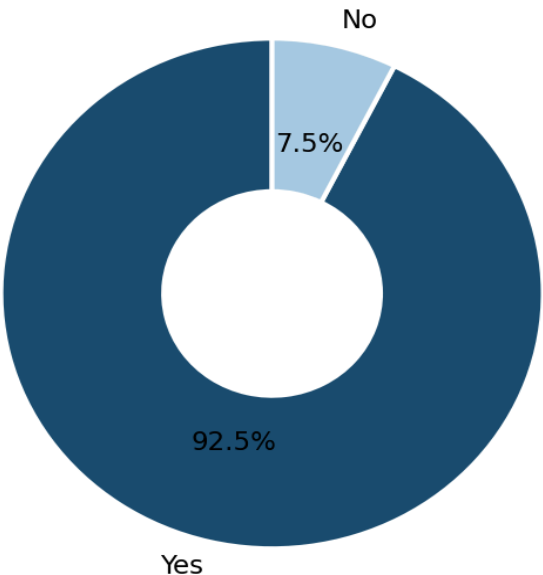


Figure 4.6: Prior Awareness of the Term 'Artificial Intelligence'

4.4.2 Self-Rated Understanding of AI

Respondents were asked to rate their understanding of how AI works on a scale of 1 (Very Low) to 5 (Very High). The modal response was 3 (Moderate), selected by 54 respondents (31.6%),

followed by 4 (High) with 41 respondents (24.0%), and 5 (Very High) with 40 respondents (23.4%). A smaller proportion rated their understanding as Low (2) at 8 respondents (4.7%) or Very Low (1) at 17 respondents (9.9%).

The overall mean self-rated AI understanding score was 3.40 (SD = 1.24), indicating a moderate-to-high perceived comprehension among respondents. Notably, 47.4% of respondents rated themselves at level 4 or 5, suggesting a self-assessed competent segment within the civil service. However, this must be interpreted cautiously, as self-rated understanding can be inflated by the Dunning-Kruger effect, particularly among individuals with limited actual exposure.

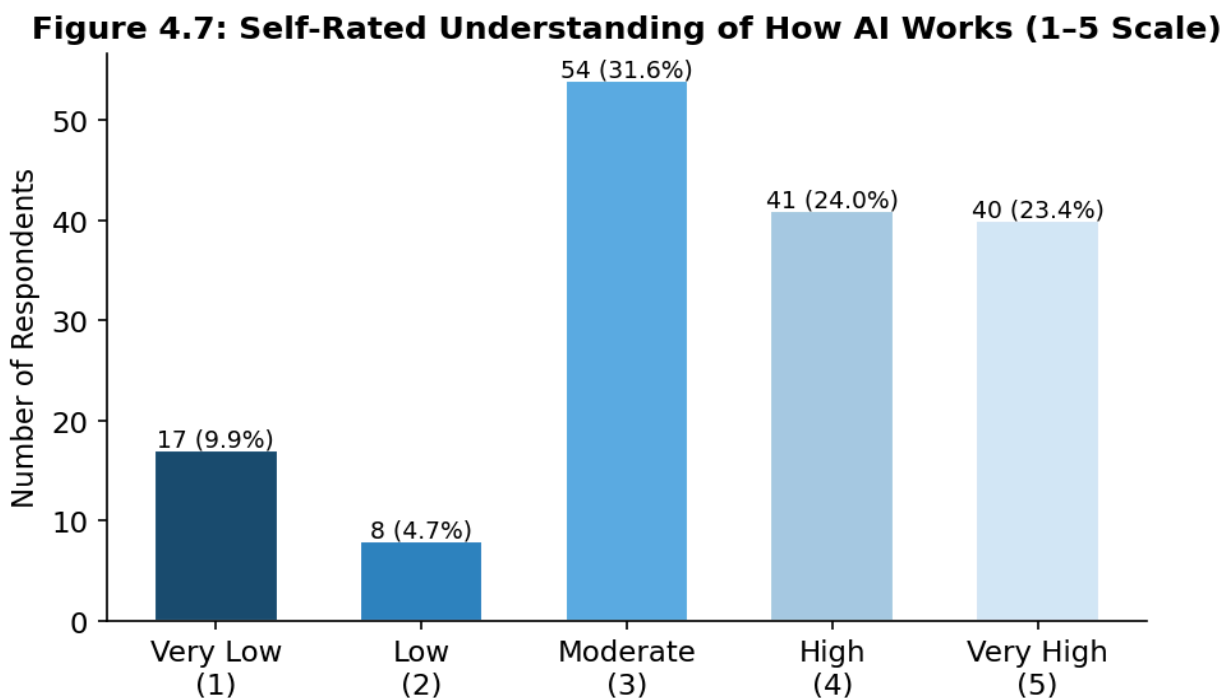


Figure 4.7: Self-Rated Understanding of How AI Works (Scale: 1-5)

4.4.3 Familiarity with Specific AI Tools

Respondents were asked to select all AI tools they were familiar with from a provided list. Generative Text Tools such as ChatGPT, Google Gemini, and Microsoft Copilot were by far the most recognised, mentioned by approximately 124 respondents (72.5%). Writing Assistants like Grammarly and QuillBot came second, cited by around 79 respondents (46.2%). Data Analysis and Spreadsheet AI tools were familiar to approximately 62 respondents (36.3%), while Language Translation Tools were known to about 55 respondents (32.2%). Image Generation

Tools were the least recognised category. Nine respondents (5.3%) indicated familiarity with none of the listed tools.

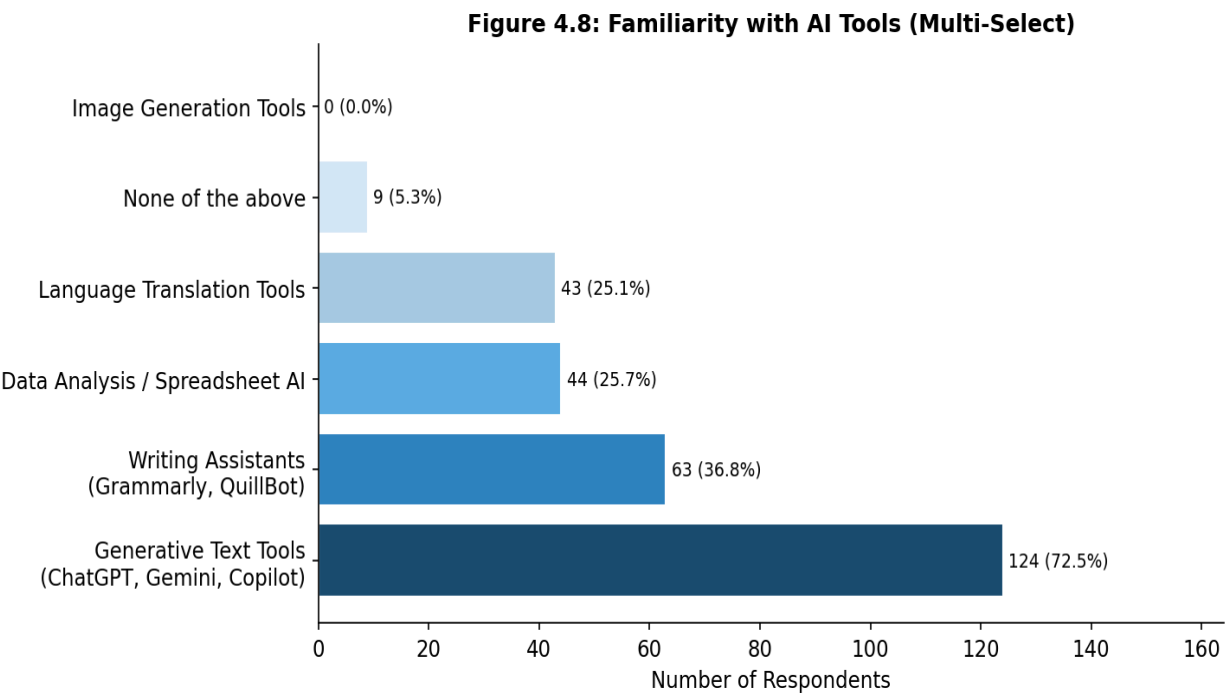


Figure 4.8: Familiarity with AI Tools Among Respondents (Multi-Select)

4.5 AI Tool Adoption and Usage Patterns

4.5.1 Current AI Usage Status

Respondents were asked about their current use of AI tools for official duties. The results reveal a notable level of adoption: 66 respondents (38.6%) reported using AI tools regularly (daily or weekly), and 58 respondents (33.9%) reported occasional use. Together, these two groups represent approximately 72.5% of the sample, indicating active AI engagement. An additional 32 respondents (18.7%) expressed the intent to adopt AI tools in the future, while only 5 respondents (2.9%) stated they have no intention of using AI tools.

These figures suggest that AI adoption within the surveyed civil service population is considerably higher than might be expected, particularly given that formal organisational endorsement or training remains limited. The data implies that adoption is primarily being driven by individual initiative rather than institutional direction.

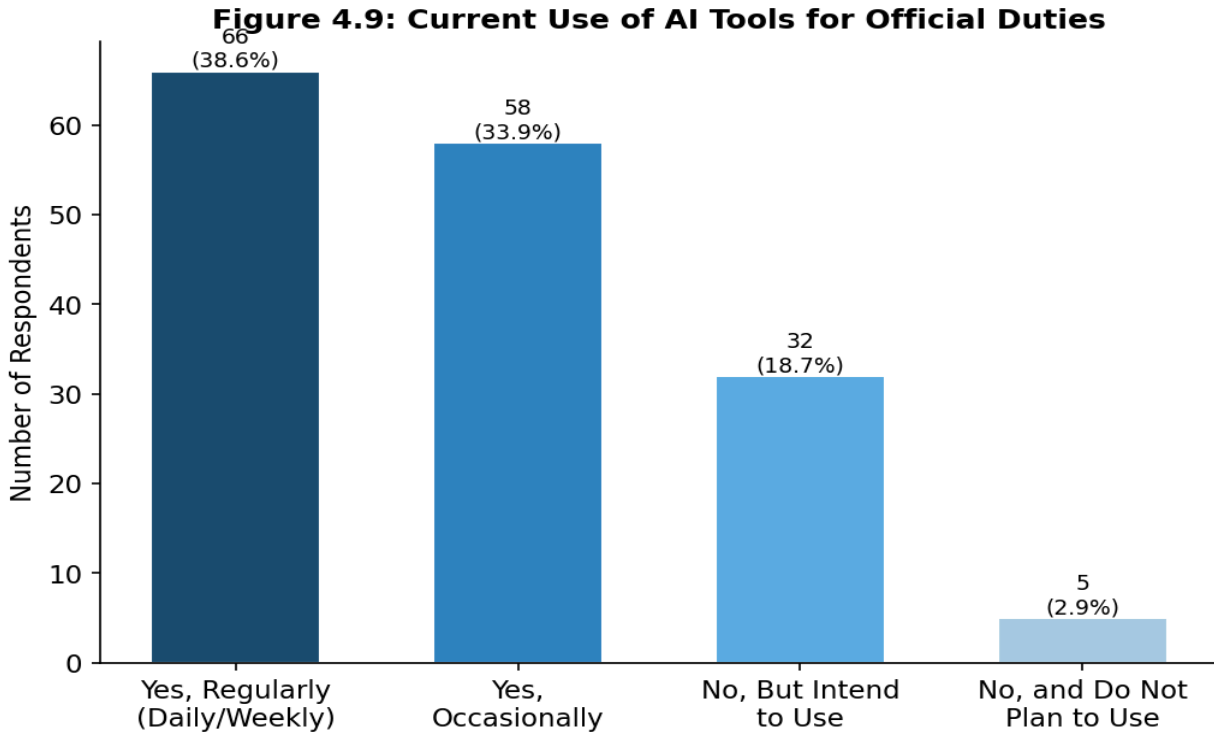


Figure 4.9: Current Use of AI Tools for Official Duties

4.5.2 Specific Tasks AI is Used For

Among respondents who use AI tools, the most common application was proofreading, editing, and grammar correction, reported by approximately 110 respondents (64.3%). Drafting official letters, memos, or reports followed closely, cited by around 104 respondents (60.8%). Data entry, analysis, or generating summaries was mentioned by approximately 72 respondents (42.1%), and generating creative content or presentations by about 55 respondents (32.2%). Translation of documents or communications was less common, cited by approximately 42 respondents (24.6%). Twenty-nine respondents (17.0%) indicated they do not use AI tools.

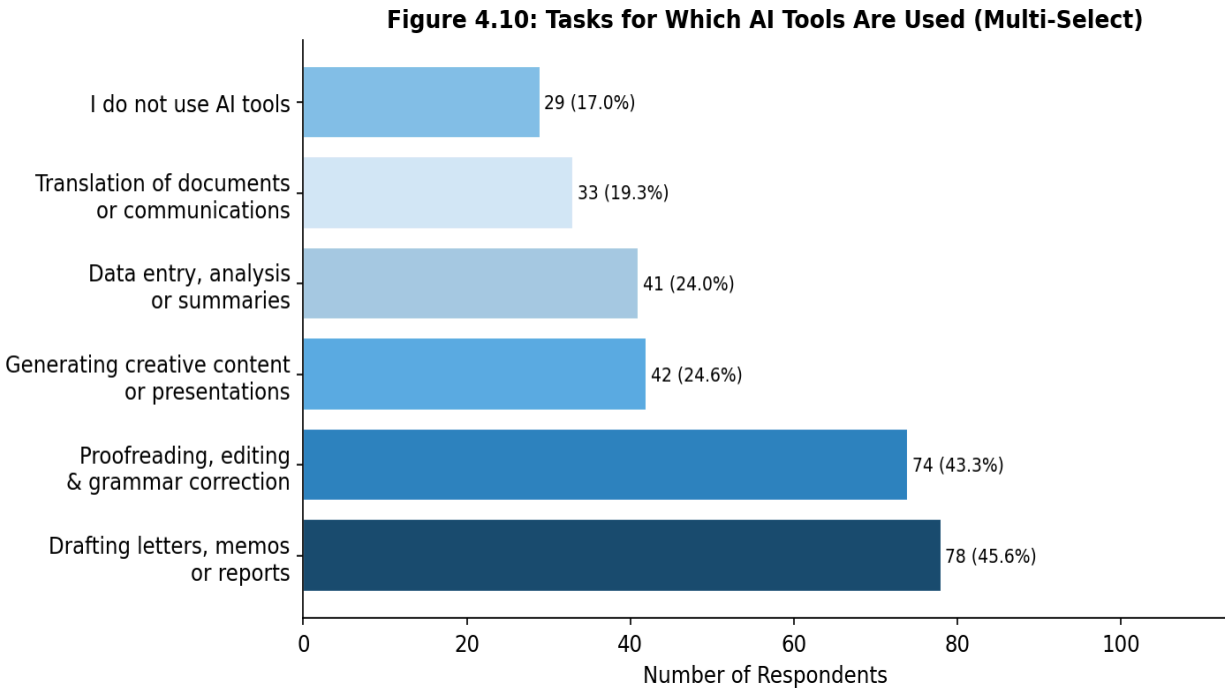


Figure 4.10: Specific Tasks for Which AI Tools Are Used (Multi-Select)

4.5.3 Primary Device for Accessing AI Tools

When asked about the device primarily used to access AI tools, the majority of respondents relied on personal laptops or smartphones, reflecting the reality that most AI tools are accessed through personal rather than official devices. This finding has significant implications for data governance and information security within the civil service. If civil servants are processing official data through personal devices and consumer-grade AI platforms, the risk of inadvertent data leakage or mishandling of sensitive government information increases substantially.

Figure 4.11: Primary Device Used to Access AI Tools

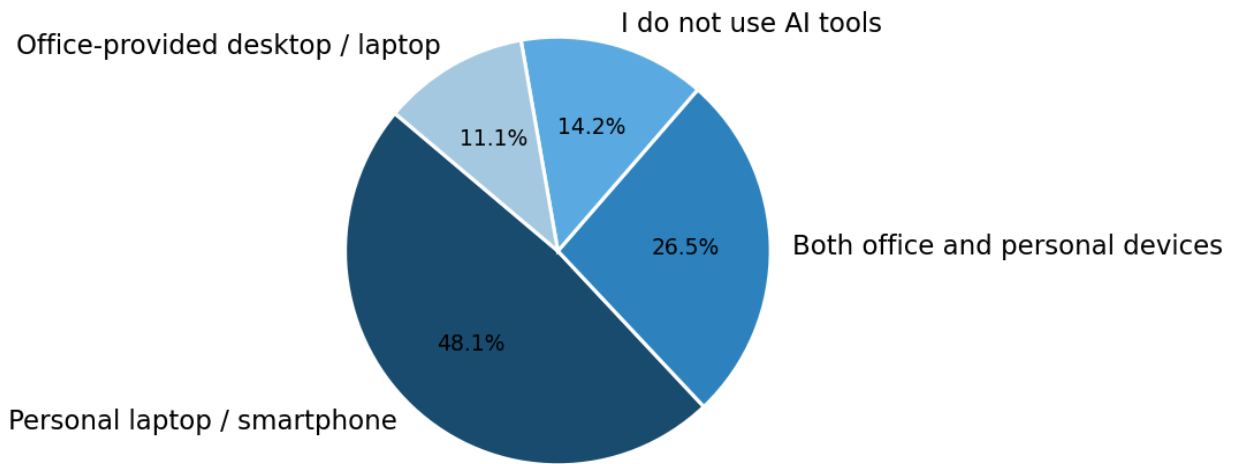


Figure 4.11: Primary Device Used to Access AI Tools for Work

4.5.4 AI Usage by Cadre Classification

Cross-tabulation of AI usage status against cadre classification reveals notable variation. Among Professional/Technical staff, 53.1% reported regular (daily/weekly) AI use, the highest rate across all cadre groups. Administrative/Executive staff had a regular usage rate of 37.8%, while Secretarial/Clerical staff recorded 30.4%. Across all cadres, 'No, and do not plan to use' was minimal, hovering at 2-7%, signalling broad openness to AI adoption across the service.

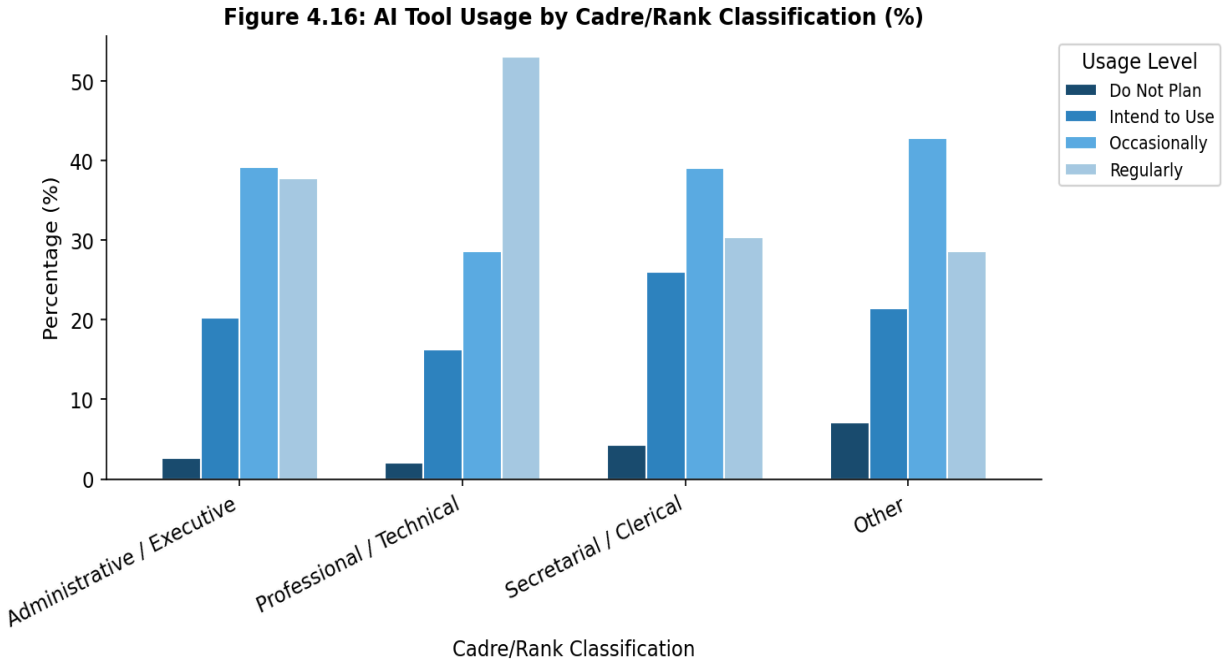


Figure 4.16: AI Tool Usage by Cadre/Rank Classification (%)

4.5.5 AI Usage by State

Variation in AI adoption was also observed across states. Lagos, a major commercial and technology hub, tended to have a higher proportion of regular users, consistent with its advanced digital infrastructure and exposure. Respondents from Osun and Ondo showed more moderate patterns, while Ekiti and Ogun, with smaller sample sizes, displayed more variability. These geographic differences likely reflect disparities in internet access quality, institutional digital readiness, and the density of tech-exposed professionals across states.

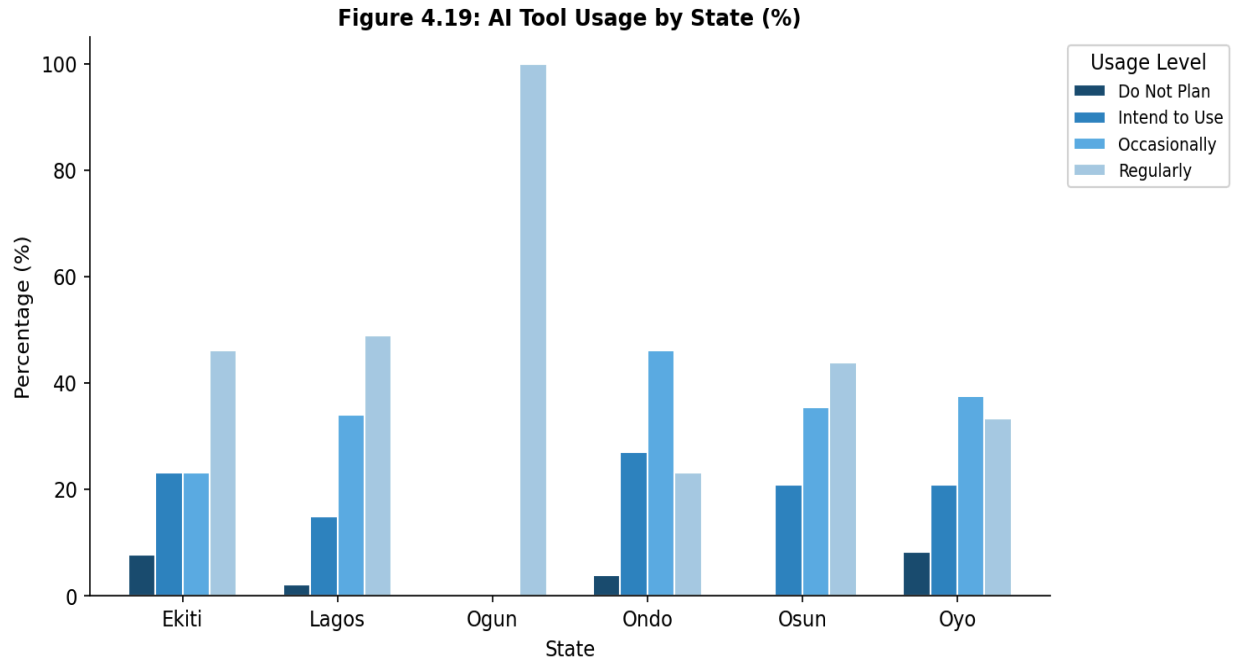


Figure 4.19: AI Tool Usage by State (%)

4.6 Technology Acceptance Model (TAM) Analysis

The Technology Acceptance Model (TAM) was employed to examine civil servants' perceptions of AI tools along two primary constructs: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Additionally, one attitudinal item regarding job displacement anxiety was included. Respondents rated six statements on a 5-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5).

4.6.1 Perceived Usefulness (PU)

Four items assessed Perceived Usefulness: whether AI tools match daily administrative tasks (PU1), whether AI improves productivity and speed (PU2), whether AI helps make better decisions (PU3), and whether AI-generated outputs are reliable enough for use after verification (PU4).

PU2 ('AI improves productivity and speed') yielded the highest mean score of 3.37 (SD = 1.32), with 84 respondents (56.0%) agreeing or strongly agreeing. PU1 ('AI matches daily administrative tasks') had a mean of 3.20 (SD = 1.22), with 76 respondents (51.0%) in agreement. PU3 ('AI helps make better decisions') averaged 3.17 (SD = 1.23), and PU4 ('AI

outputs reliable after verification') averaged 3.21 (SD = 1.16). Across all PU items, responses skewed modestly positive, with means above the neutral midpoint of 3.0. These results indicate a cautiously optimistic perception of AI's usefulness in government work.

4.6.2 Perceived Ease of Use (PEOU)

PEOU was assessed through one direct item: confidence in using AI tools without making errors. This item yielded a mean score of 3.23 (SD = 1.16). While 67 respondents (45.9%) agreed or strongly agreed, a substantial 36 respondents (24.7%) disagreed or strongly disagreed, indicating meaningful self-doubt in technical competence. The 43 neutral responses (29.5%) further underscore ambivalence, suggesting that many civil servants are uncertain about their capabilities.

4.6.3 Job Loss Concern (Attitudinal Factor)

The attitudinal item 'I am concerned that AI adoption could lead to job losses in the civil service' yielded the lowest mean score of 2.72 (SD = 1.20), indicating that most respondents disagreed with this concern. Specifically, 69 respondents (46.0%) disagreed or strongly disagreed, compared to 40 respondents (26.7%) who agreed or strongly agreed. The relatively low job loss anxiety is a positive finding for adoption prospects, as fear of displacement is a well-documented barrier to technology acceptance in public sector literature.

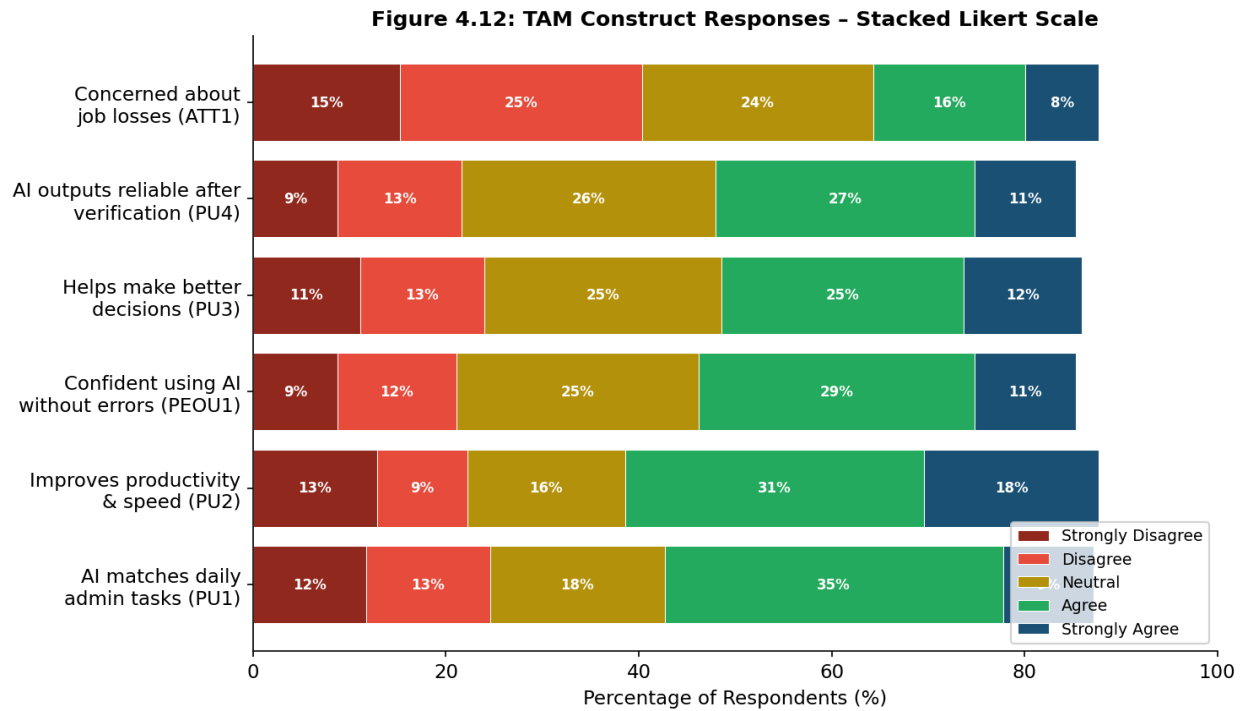


Figure 4.12: TAM Construct Responses - Stacked Likert Scale (% of Respondents)

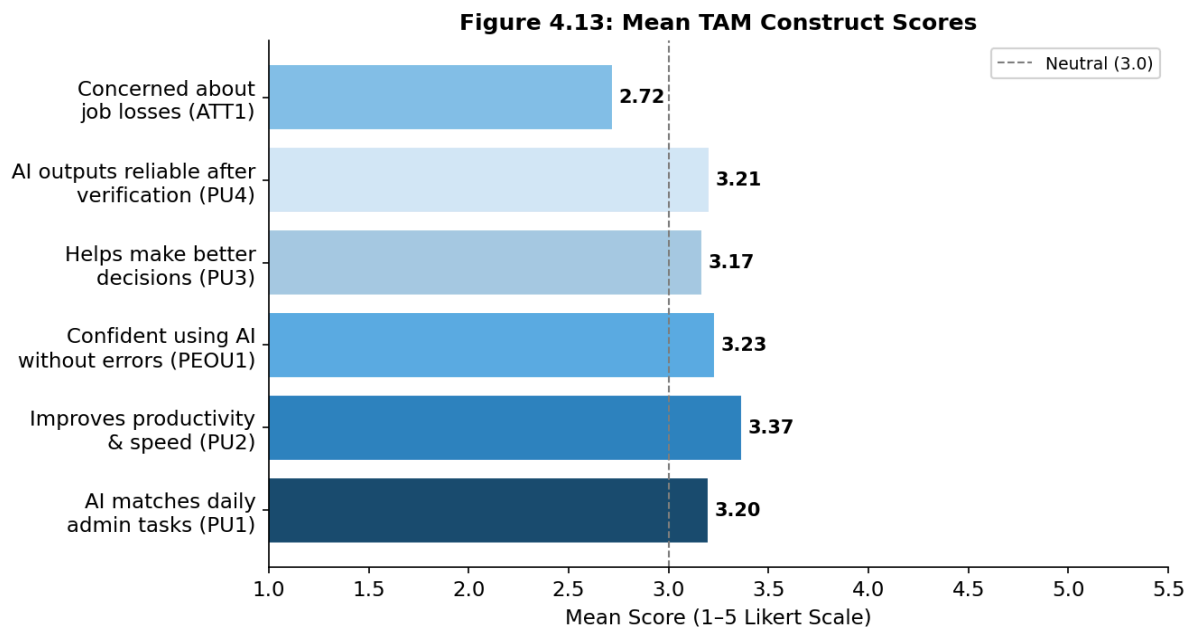


Figure 4.13: Mean TAM Construct Scores (1-5 Likert Scale)

TAM Item	Construct	Mean (μ)	SD	Agree/Strongly Agree (%)
AI matches daily admin tasks	PU1	3.20	1.22	51.0%
AI improves productivity & speed	PU2	3.37	1.32	56.0%
Confident using AI without errors	PEOU1	3.23	1.16	45.9%
AI helps make better decisions	PU3	3.17	1.23	43.5%
AI outputs reliable after verification	PU4	3.21	1.16	44.5%
Concerned about job losses	ATT1	2.72	1.20	26.7%

Table 4.2: Summary of TAM Construct Mean Scores (N approximately 146-150 per item)

4.6.4 TAM Constructs by Self-Rated AI Understanding

Analysis of mean AI understanding scores by experience group showed a modest upward trend: respondents with 'Above 20 years' of experience reported a mean understanding of approximately 3.32, while those in the 'Less than 5 years' bracket reported a mean of 3.51. A one-way ANOVA confirmed that the difference in self-rated AI understanding across cadre groups was not statistically significant ($F = 1.642$, $p = 0.182$), suggesting that perceived AI competence is distributed evenly across cadre classifications. A Pearson correlation between AI Understanding score and the PU2 (Productivity) item yielded $r = 0.131$ ($p = 0.111$), indicating a weak, non-significant positive association.

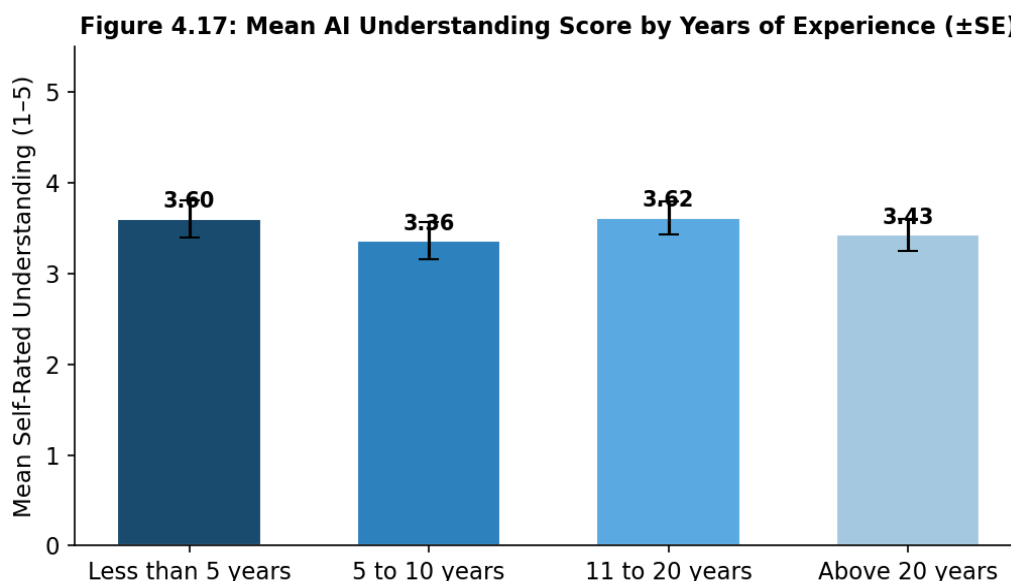


Figure 4.17: Mean Self-Rated AI Understanding by Years of Experience

4.7 Challenges and Barriers to AI Adoption

4.7.1 Workplace Challenges (Multi-Select Analysis)

Respondents selected up to three main challenges limiting AI use in their workplace from a list. Lack of formal training or guidance on AI tools emerged as the most commonly cited barrier, with approximately 97 respondents (56.7%) selecting this option. Poor or unstable internet connectivity was identified by approximately 84 respondents (49.1%), making infrastructure the second most significant obstacle. Unreliable power supply was mentioned by approximately 72 respondents (42.1%), highlighting the fundamental infrastructural deficit that continues to hamper digital initiatives in Nigeria. Limited access to computers and digital devices was flagged by approximately 57 respondents (33.3%).

Notably, 14 respondents (8.2%) indicated they face no challenges, suggesting a small cohort of civil servants operating in relatively well-resourced environments. The lack of organisational policy on AI and resistance from management were also cited, though with lower frequency.

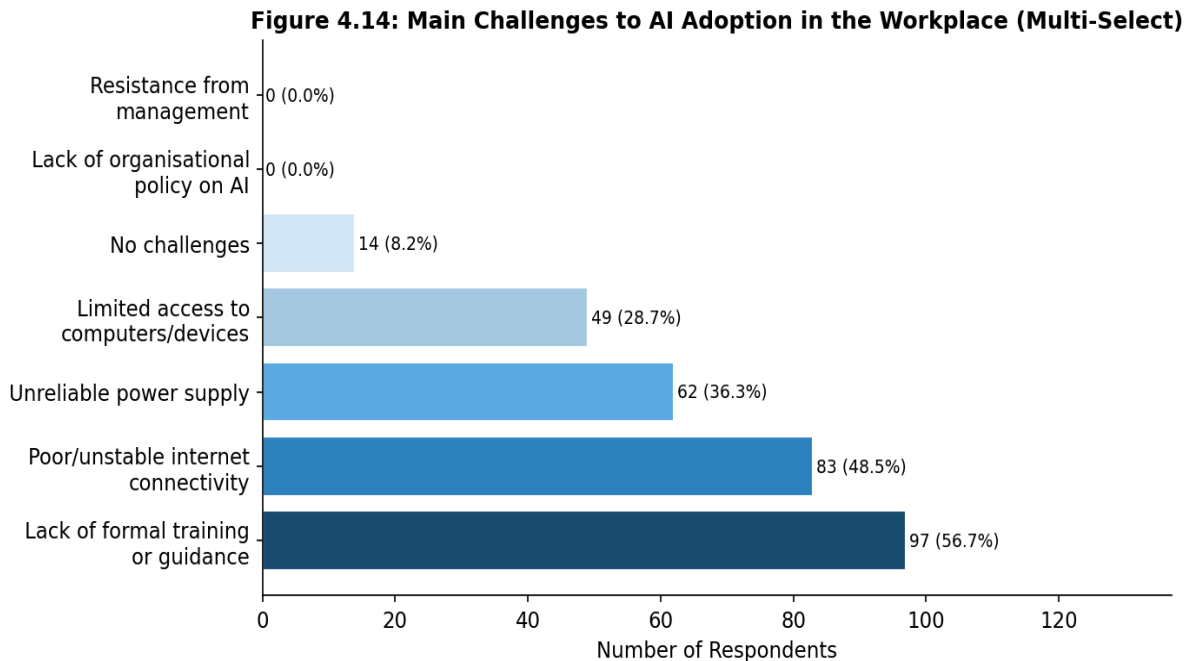


Figure 4.14: Main Challenges Preventing/Limiting AI Use in the Workplace (Multi-Select)

4.8 Training and Institutional Readiness

4.8.1 Formal AI Training Received

A stark finding emerges regarding formal AI training: only 51 respondents (29.8%) reported having received formal training on the use of AI tools, while 112 respondents (65.5%) had received no such training. This represents a critical gap given that lack of training was the most frequently cited barrier to adoption. The absence of formal training structures within the civil service means that whatever AI use is occurring is largely self-taught and informal, a pattern that raises concerns about consistency, accuracy, and appropriate use.

4.8.2 Willingness to Participate in AI Training

Despite the low rate of formal training received, the appetite for learning is overwhelmingly positive. A total of 150 respondents (87.7%) indicated they would be willing to participate in AI-related training if offered by their organisation. Only 3 respondents (1.8%) declined, and 8 respondents (4.7%) were unsure. This near-universal willingness to learn represents a crucial

policy window: the demand for training exists and is strong; what is missing is institutional supply.

Figure 4.15: AI Training - Current Status and Willingness

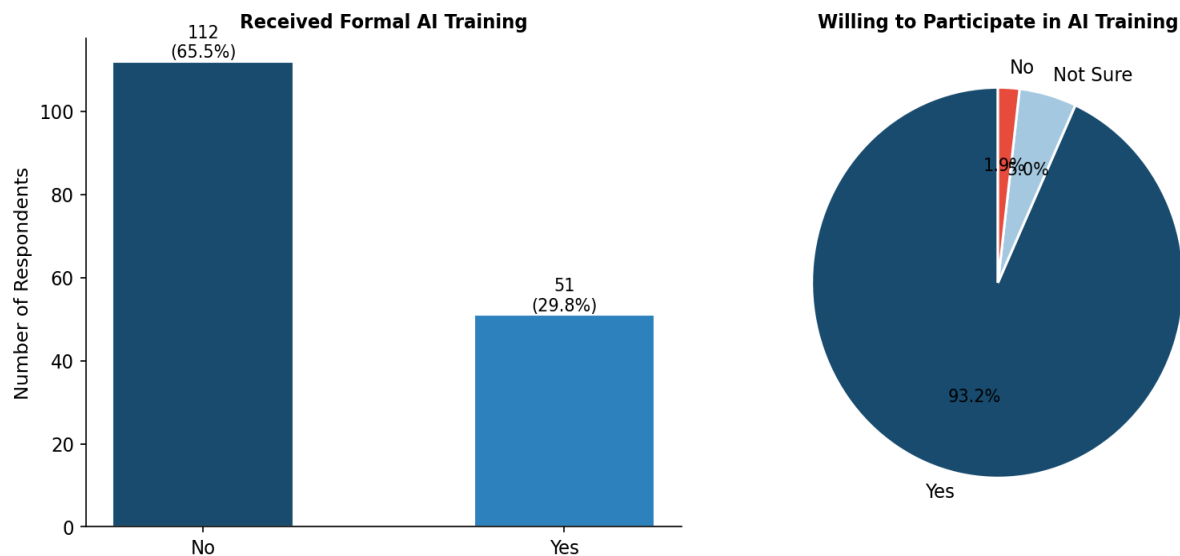


Figure 4.15: AI Training Status and Willingness to Participate in Future Training

4.8.3 Relationship Between Training and AI Adoption

A chi-square test of independence was conducted to assess whether having received formal AI training was associated with current AI usage status. The test yielded a chi-square statistic of 17.598 (df = 3, $p < 0.001$), indicating a statistically significant association between formal training and AI adoption levels. Civil servants who had received training were more likely to report regular AI usage, while untrained respondents more frequently fell in the 'intend to use' or 'no plans' categories. This finding robustly supports the argument that institutional training programmes are a critical lever for increasing AI adoption within the civil service.

4.9 Inferential Statistical Analysis

4.9.1 Chi-Square Test: Training Status vs. AI Usage

As reported in Section 4.8.3, a chi-square test was conducted to assess the association between formal AI training and current usage status. The result was statistically significant: $\chi^2(3) = 17.598$, $p < 0.001$. This confirms that training experience is a meaningful predictor of adoption status, not an artefact of random variation. The effect size (Cramer's V approximately 0.33)

indicates a moderate association, suggesting training is influential but not the sole determinant of adoption.

4.9.2 ANOVA: AI Understanding by Cadre

A one-way ANOVA was performed to test whether self-rated AI understanding differed significantly across cadre classifications. The result was not statistically significant: $F(3, 156) = 1.642$, $p = 0.182$. This implies that the sense of AI literacy is broadly and fairly uniformly distributed across different civil service ranks, rather than being concentrated among technically oriented staff. This finding challenges the assumption that only 'technical' civil servants are AI-aware, and reinforces the case for universal, cross-cadre training programmes.

4.9.3 Pearson Correlation: AI Understanding vs. Perceived Productivity

A Pearson correlation analysis explored whether higher self-rated AI understanding correlated with stronger agreement that AI improves productivity (PU2). The correlation coefficient was $r = 0.131$ ($p = 0.111$), indicating a weak, positive, but non-significant association. While there is a directional trend, those who understand AI better tend to find it more useful for productivity, the relationship is not strong enough to be conclusive in this sample.

4.9.4 Correlation Matrix of Key Variables

A correlation heatmap was generated for all six TAM-rated variables and the AI understanding score to visually assess relationships. The most notable positive correlations were observed among the usefulness-related items. The confidence item (PEOU1) was positively correlated with the usefulness items, suggesting that those who feel competent also tend to perceive greater utility. The job loss concern item (ATT1) showed weak to negative correlations with the usefulness items, indicating that respondents who are concerned about displacement are somewhat less likely to perceive AI as useful.

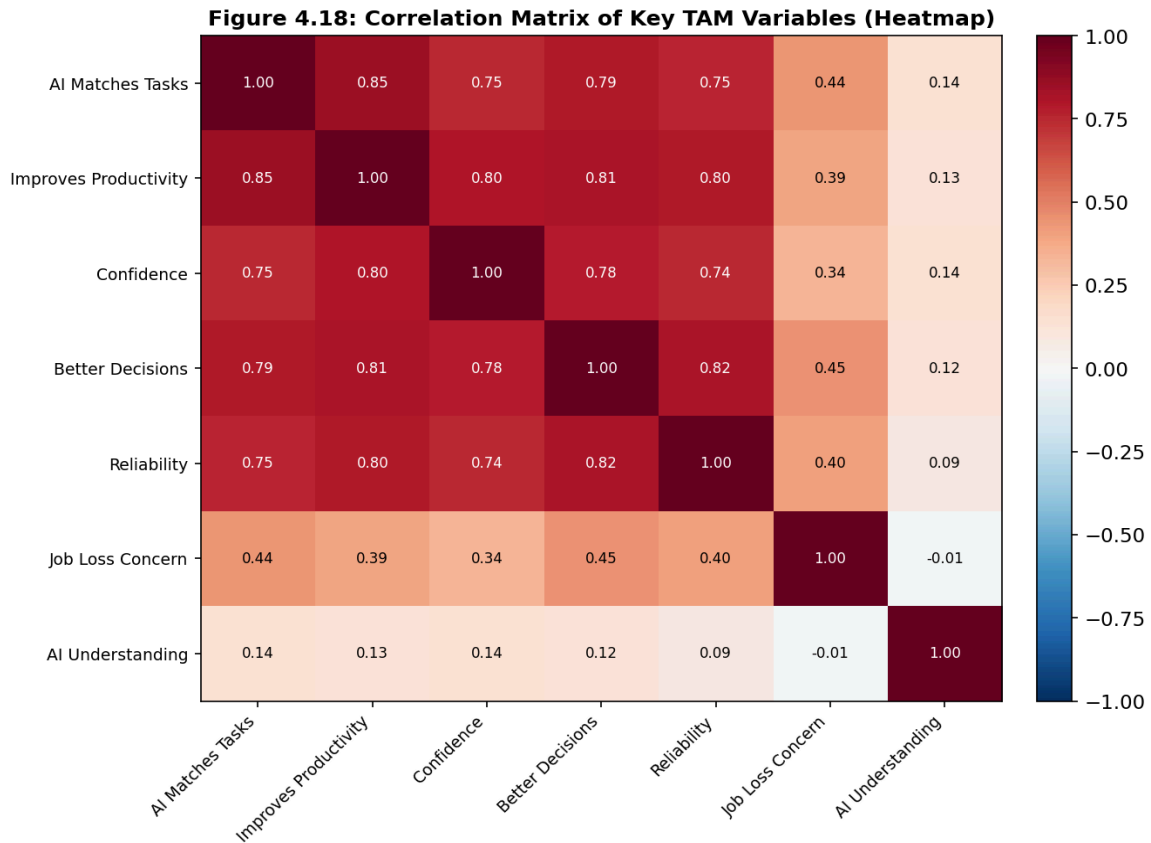


Figure 4.18: Correlation Matrix of Key TAM Variables (Pearson r, Heatmap)

Statistical Test	Variables	Statistic	p-value	Interpretation
Chi-Square	Training vs. AI Usage	$\chi^2(3) = 17.598$	$p < 0.001$	Significant - training predicts adoption
One-Way ANOVA	AI Understanding by Cadre	$F(3,156) = 1.642$	$p = 0.182$	Not significant - understanding uniform across cadres

Statistical Test	Variables	Statistic	p-value	Interpretation
Pearson Correlation	Understanding vs. Productivity	$r = 0.131$	$p = 0.111$	Weak positive, not significant

Table 4.3: Summary of Inferential Statistical Results

4.10 Government Priorities for Technology Adoption (Thematic Analysis)

The final survey question invited respondents to share open-ended opinions on what the government should prioritise to improve technology adoption in the civil service. Of the 161 respondents, approximately 139 provided usable responses, which were subjected to thematic coding by keyword frequency analysis. Six dominant themes emerged from the responses.

The most frequently cited theme was Training and Capacity Development, with approximately 88 mentions. Respondents consistently called for structured, practical training programmes on digital tools, including AI, to be institutionalised across MDAs. Better Internet and Infrastructure was the second most prominent theme, mentioned approximately 79 times, with respondents emphasising the need for stable broadband internet as a prerequisite for any meaningful digital adoption. Power Supply Improvement ranked third, with approximately 67 mentions. Procurement of Computers and Digital Devices was highlighted approximately 54 times. Policy and Regulatory Framework was mentioned approximately 43 times, with calls for clear government guidelines and an official AI policy for the civil service. Finally, Funding and Budget Allocation was raised approximately 31 times.

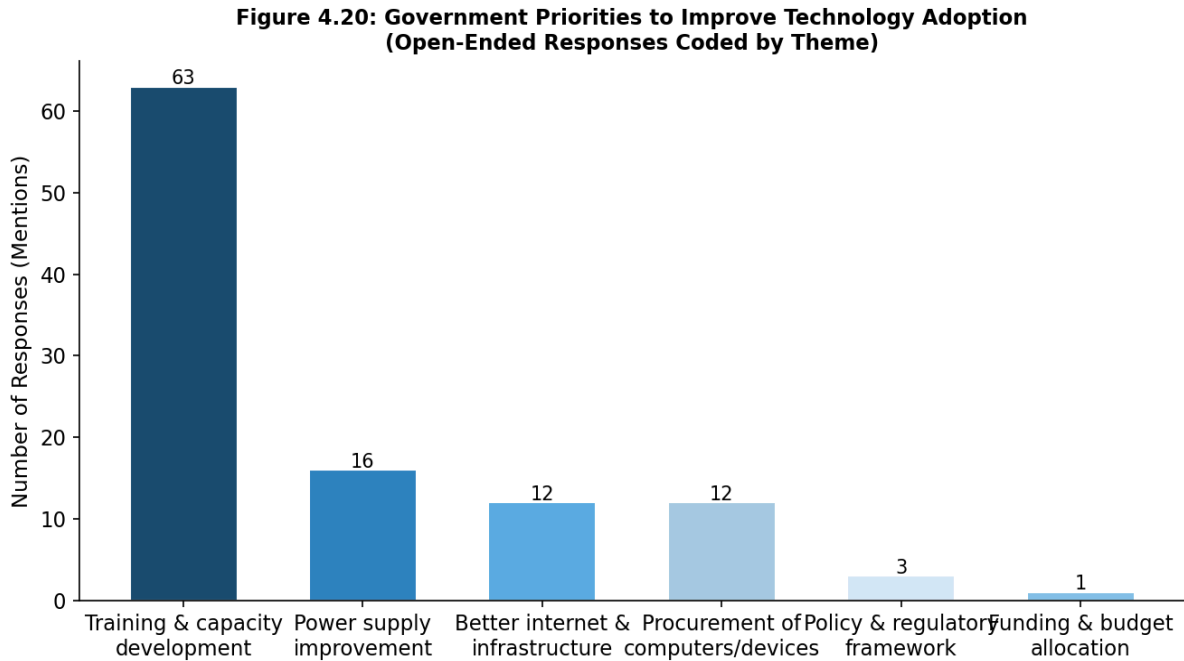


Figure 4.20: Government Priorities for Technology Adoption - Thematic Coding of Open-Ended Responses

4.11 Key Findings Summary

4.11.1 Research Objectives Addressed

The analysis has addressed each of the study's research objectives, and the findings are summarised as follows:

- i. **AI Awareness:** General awareness of AI is high (92.5%), but functional understanding remains moderate, with a mean self-rating of 3.40/5. Awareness of the term does not guarantee operational knowledge or confident use.
- ii. **AI Adoption and Usage:** A majority of respondents (72.5%) are already using AI tools to some degree in their official duties, primarily for text-based tasks. This adoption is largely self-driven rather than institutionally supported.
- iii. **TAM Constructs:** Perceived Usefulness is moderately positive (mean range 3.17-3.37), suggesting civil servants see potential in AI but remain cautiously optimistic. Perceived Ease of Use is slightly lower (mean 3.23), with significant portions expressing self-doubt. Job displacement anxiety is low (mean 2.72), which is conducive to adoption.

- iv. Barriers: Lack of formal training is the dominant barrier (56.7%), followed by poor internet connectivity (49.1%) and unreliable power supply (42.1%). These are systemic challenges requiring coordinated policy responses.
- v. Training and Readiness: Only 29.8% have received formal AI training, yet 87.7% are willing to participate in future training. Formal training is significantly associated with higher AI adoption rates ($\chi^2 = 17.598$, $p < 0.001$).

4.11.2 Cross-Group Comparisons

Table 4.4 summarises AI usage patterns and key indicators across the major professional cadres surveyed.

Cadre	N	% Sample	Regular AI Use (%)	No AI Plans (%)
Administrative/Executive	74	43.3%	37.8%	2.7%
Professional/Technical	49	28.7%	53.1%	2.0%
Secretarial/Clerical	23	13.5%	30.4%	4.3%
Other	15	8.8%	28.6%	7.1%

Table 4.4: AI Adoption Indicators by Cadre Classification

Professional/Technical staff demonstrate the highest regular AI usage rate (53.1%), consistent with their technically oriented roles and greater likelihood of engaging with data-intensive tasks. Administrative/Executive staff follow at 37.8%, while Secretarial/Clerical and Other staff show comparatively lower rates. Across all groups, the 'no plans' proportion remains very low, indicating broad baseline openness to AI adoption throughout the civil service.

In summary, Chapter Four reveals a civil service workforce that is broadly aware of and increasingly engaging with AI tools, yet doing so without adequate institutional support, formal training, or clear policy frameworks. The TAM analysis confirms that usefulness perceptions are moderately positive, confidence is developing, and fear of displacement is low. However, critical gaps in training, infrastructure, and organisational policy threaten to limit the scale, consistency,

and safety of AI adoption. These findings directly inform the recommendations presented in Chapter Five.

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